

A cubic-box counterexample to Conjecture 425

Local AI-assisted solution draft

3 October 2026

Source and interpretation. Conjecture 00000000425 (original statement and pinned commit in `SOURCE.md`) asserts that the coefficient sequence in the diagonal specialization of the boxed plane partition generating function is always log-concave. We use the standard volume generating polynomial, with the diagonal specialization meaning equal box side lengths. The counterexample is the $2 \times 2 \times 2$ box.

Definitions. A plane partition in this box is an array $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with entries in $\{0, 1, 2\}$, weakly decreasing along both rows and columns: $a \geq b$, $a \geq c$, $b \geq d$, and $c \geq d$. Its volume is $a + b + c + d$. Let c_k count such arrays of volume k . Log-concavity requires $c_k^2 \geq c_{k-1}c_{k+1}$ at every interior index.

Counterexample and proof. There is exactly one array of volume zero, namely the zero array, and exactly one of volume one, with a single 1 in the top-left entry. The three arrays of volume two are

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

These are exhaustive: either one entry equals 2, which by monotonicity must be the top-left entry, or two entries equal 1, giving exactly the two displayed order ideals. Consequently

$$c_0 = 1, \quad c_1 = 1, \quad c_2 = 3, \quad c_1^2 = 1 < 3 = c_0c_2.$$

Thus the asserted universal log-concavity is false. It is unnecessary to consider the additional equality characterization after this first clause has failed.

Independent polynomial check. The complete polynomial, obtained both by direct enumeration and by the MacMahon product, is

$$\prod_{i,j,k=1}^2 \frac{1 - q^{i+j+k-1}}{1 - q^{i+j+k-2}} = 1 + q + 3q^2 + 3q^3 + 4q^4 + 3q^5 + 3q^6 + q^7 + q^8.$$

The accompanying Python verifier independently computes the product as an exact integer formal power series and enumerates the arrays.

Direct formal connection to the MacMahon product. Write its numerator and denominator as $N(q)$ and $D(q)$. For the cubic box of side two, the exact triple products give

$$D(q) = 1 - q - 3q^2 + O(q^3), \quad N(q) = 1 - q^2 + O(q^3).$$

Since $D(0) = 1$, formal division defines a coefficient sequence f with $D(q)F(q) = N(q)$. Comparing the first three coefficients gives $f_0 = 1$, $f_1 - f_0 = 0$, and $f_2 - f_1 - 3f_0 = -1$; thus again $(f_0, f_1, f_2) = (1, 1, 3)$. This directly refutes log-concavity for the product specified in the conjecture, without using the counting formula as an unproved assumption.

The self-contained Lean 4.19.0 file `Main.lean` formalizes both routes. It verifies the complete, duplicate-free enumeration of the 81 arrays and the precise plane-partition predicate. It also defines integer formal series, Cauchy multiplication, and the actual three nested MacMahon products. The terminating recursion `formalQuotient` constructs every coefficient; `formalQuotient_spec` proves $DF = N$ at every index whenever $D(0) = 1$. The theorem `macmahon_product_counterexample` therefore refutes log-concavity for an explicitly constructed infinite coefficient sequence, not for a predicate whose existence is merely assumed. Its first three coefficients are also proved equal to the direct enumeration above. All calculations use kernel proofs, with no placeholders or custom axioms.

Alternative reading of “coefficientwise.” If the intended claim instead compares consecutive polynomials $F_n(q)$ for $n \times n \times n$ boxes, it is false as well. The completely filled box is the unique term of maximum volume, so F_n is monic of degree n^3 . At $n = 2$, the polynomial $F_2(q)^2 - F_1(q)F_3(q)$ has coefficient -1 at q^{28} , because the two product degrees are 16 and 28. This additional argument covers that reading in ordinary mathematics; the Lean artifact formalizes the scalar coefficient-sequence reading above. The source does not explicitly define “diagonal specialization,” so we retain this interpretation boundary rather than claiming all possible meanings have been formalized.